

Multiple Choice (10 marks)

1. Which animals would most likely be helped by flood in a coastal area?
 - (a) deer
 - (b) raccoons
 - (c) alligators
 - (d) coyotes
2. How does acid rain most likely affect forests?
 - (a) Top soil loses all of its nutrients.
 - (b) Plants begin to grow deeper roots.
 - (c) Animals have more food sources.
 - (d) Trees become less healthy over time.
3. Giant redwood trees change energy from one form to another. How is energy changed by the trees?
 - (a) They change chemical energy into kinetic energy.
 - (b) They change solar energy into chemical energy.
 - (c) They change wind energy into heat energy.
 - (d) They change mechanical energy into solar energy.
4. Two layers of sedimentary rock are exposed on the side of a hill. Only one of the layers has fossils. The lack of fossils in one layer of sedimentary rock is most likely due to
 - (a) changing environments.
 - (b) volcanic activity.
 - (c) changing rates of erosion.
 - (d) evaporation of sea water.
5. Although they belong to the same family, an eagle and a pelican are different. What is one difference between them?
 - (a) their preference for eating fish
 - (b) their ability to fly
 - (c) their method of reproduction
 - (d) their method of catching food

True / False (10 marks)

Answer True or False

6. True or False: The mass of the feathers is the same as the mass of the iron nails when comparing equal volumes of both.
7. True or False: Grass gets its energy from sunlight through the process of photosynthesis.
8. True or False: Moons are celestial bodies that can exist both inside and outside the solar system.
9. True or False: The moon is a source of light because it reflects sunlight.
10. True or False: The respiratory system has layers of smooth muscle that contract to circulate blood and nutrients throughout the body.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that influence the decomposition of organic materials in nature and analyze which items would decompose the fastest and why.
12. Explain the processes involved in the transformation of sedimentary rock into metamorphic rock, discussing the conditions and factors that contribute to this change.

End of Paper